Supplementary Figures

Fig. S1 | *tbd* Map of 35 islands surveyed by CREP across the tropical Pacific.

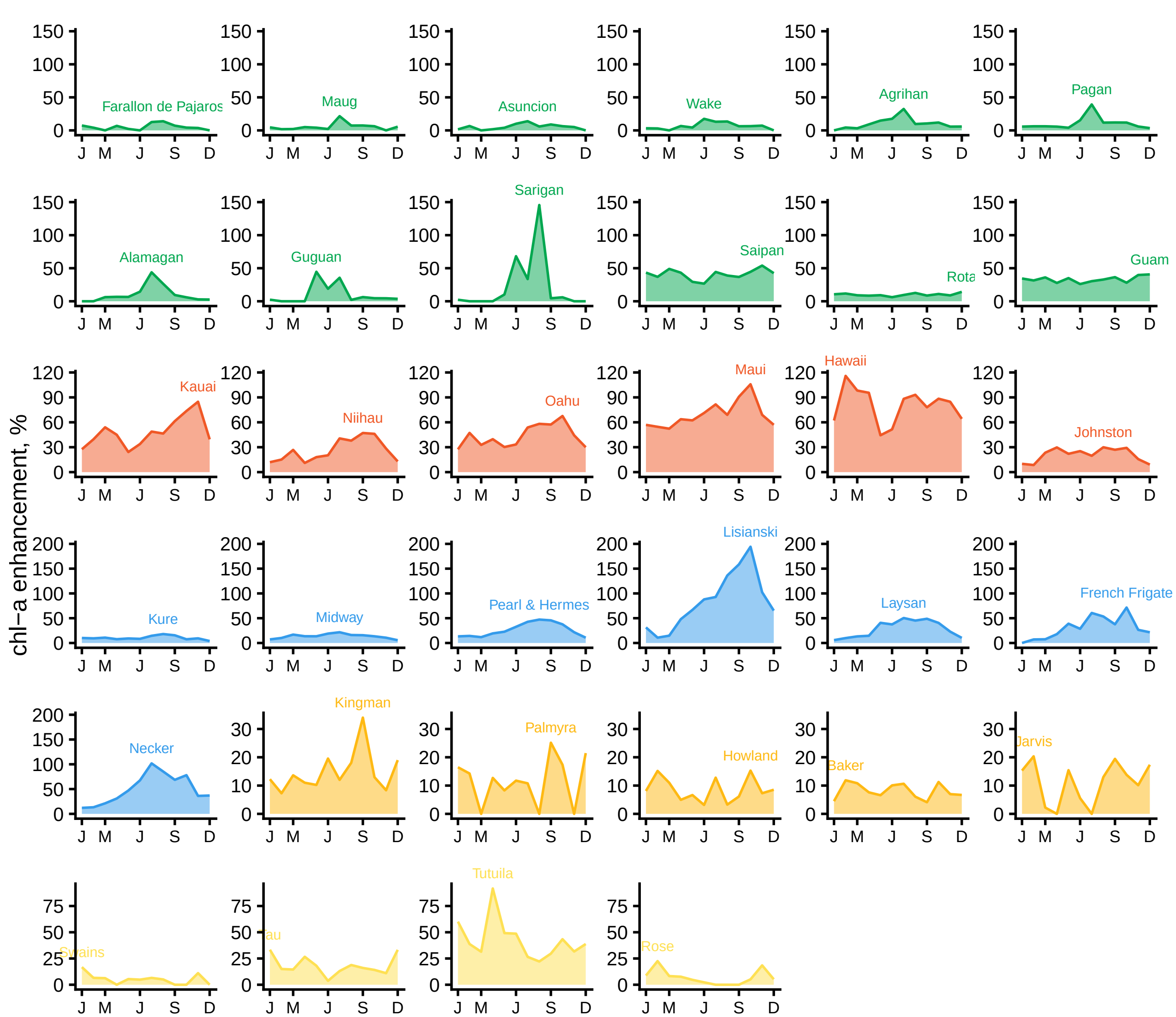

Fig. S2 | Seasonal variation in IME strength at each island. Lines are mean-monthly chl-a enhancement from January to December, ordered and coloured by island group.

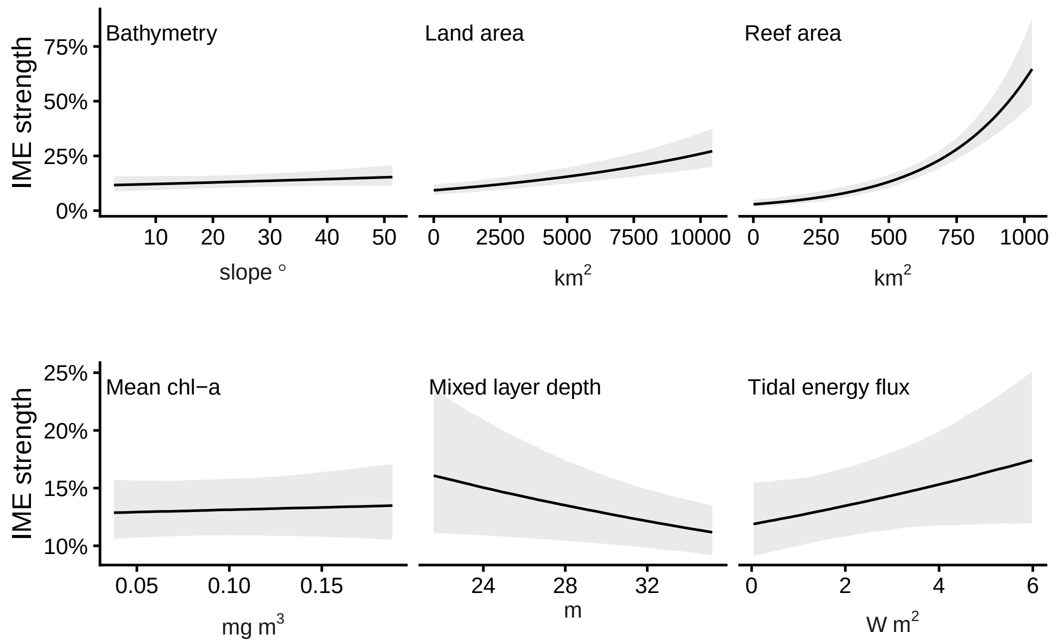

Fig. S3 | Spatial effects on chl-a enhancement for geomorphic features (top) and average oceanographic conditions (bottom) Lines are median posterior predictions (± 95% certainty intervals). Note different y-axis values between static and dynamic covariates.

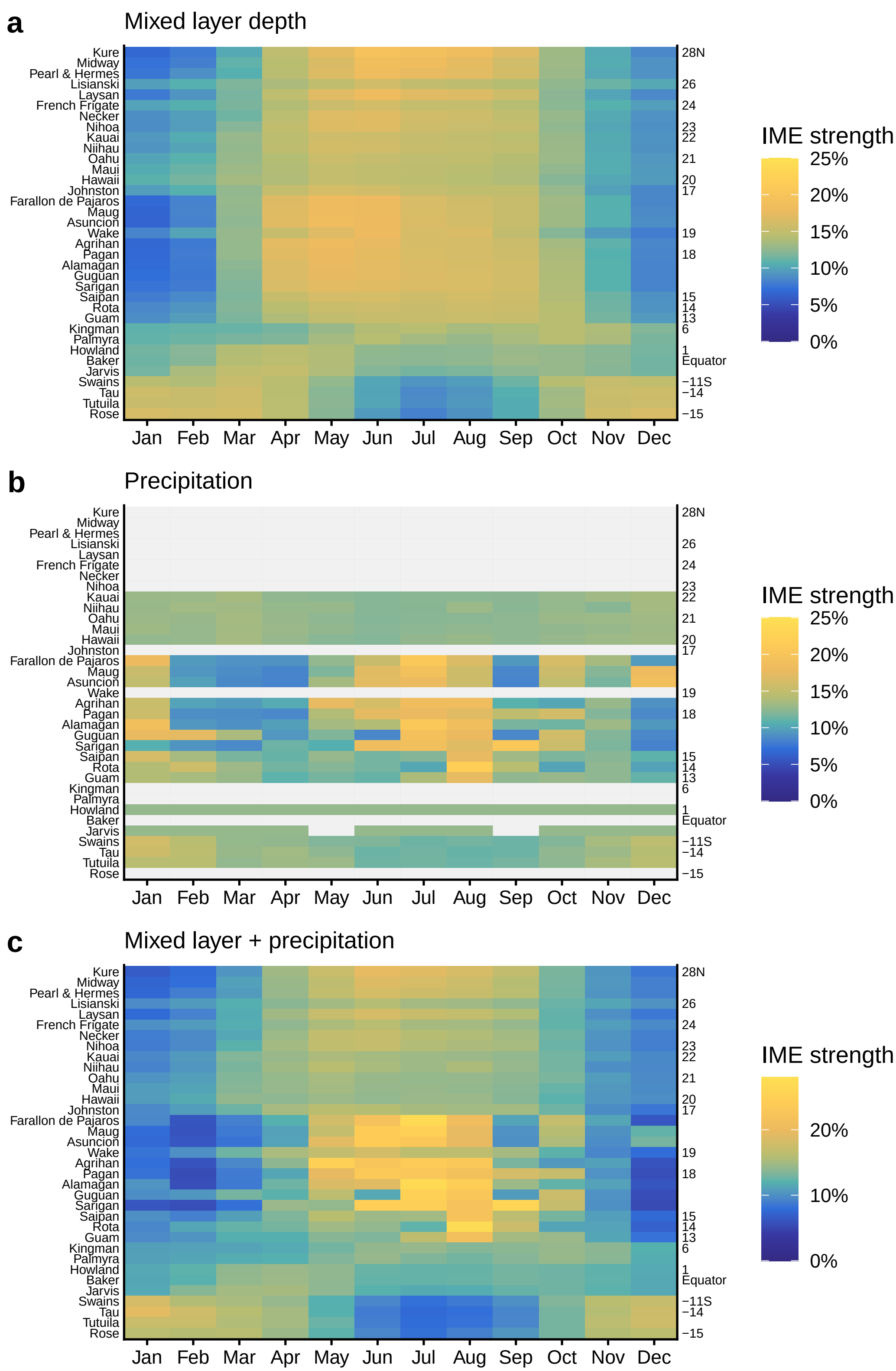

Fig. S4 | Heatmap of seasonal effects on chl-a enhancement for a mixed layer depth, b precipitation and c both covariates combined. Islands with less than 1 mm monthly rainfall are shown in grey. All values are median posterior predictions excluding effects of other covariates.

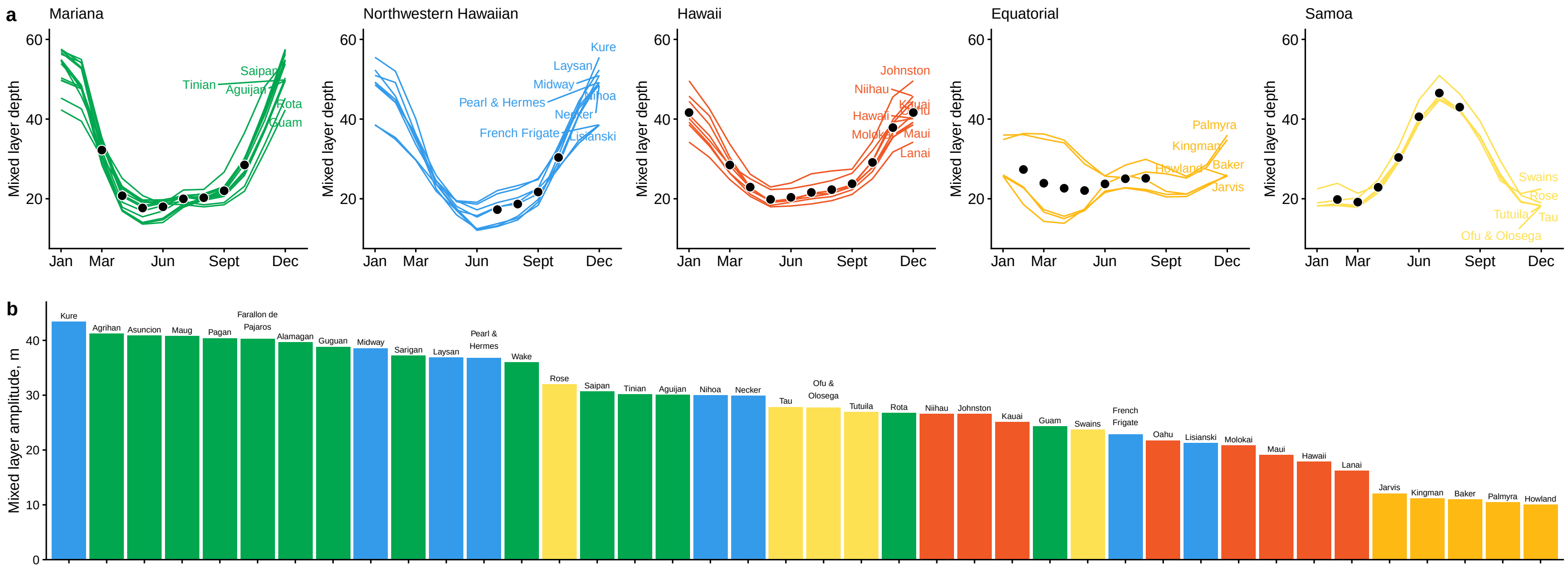

Fig. S5 | Seasonal variability in the mixed layer depth at each island. In a, lines are predicted mixed layer depth at each island from a General Additive Model with smoothers for time and month (cyclic), overlaid with points indicating timing of coral reef fish surveys.

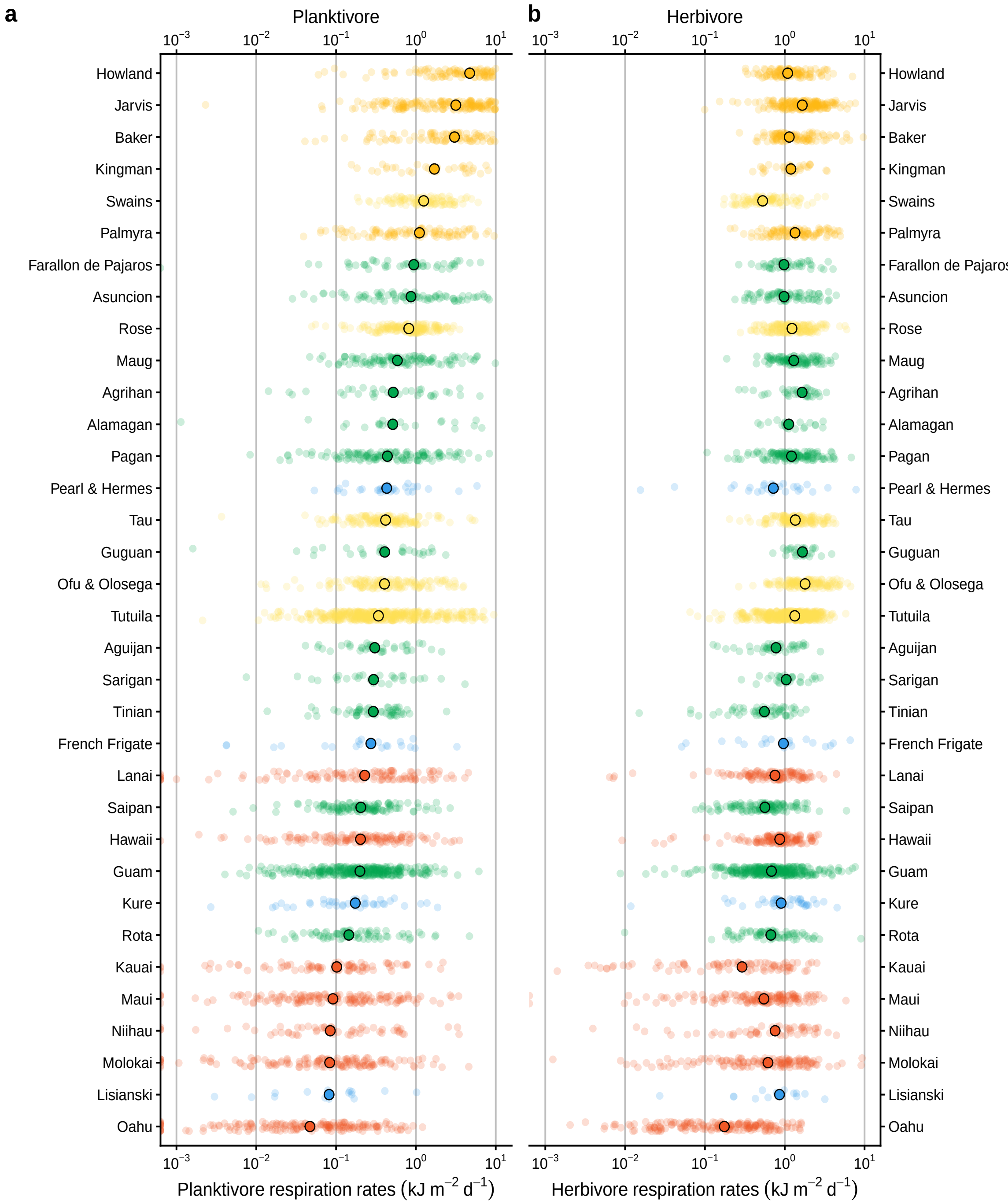

Fig. S6 | Respiration rate of reef-associated planktivores on coral reefs at 34 Pacific islands and atolls. Each point is the median respiration at each island, with jitter showing unique sites (n = 3,622).

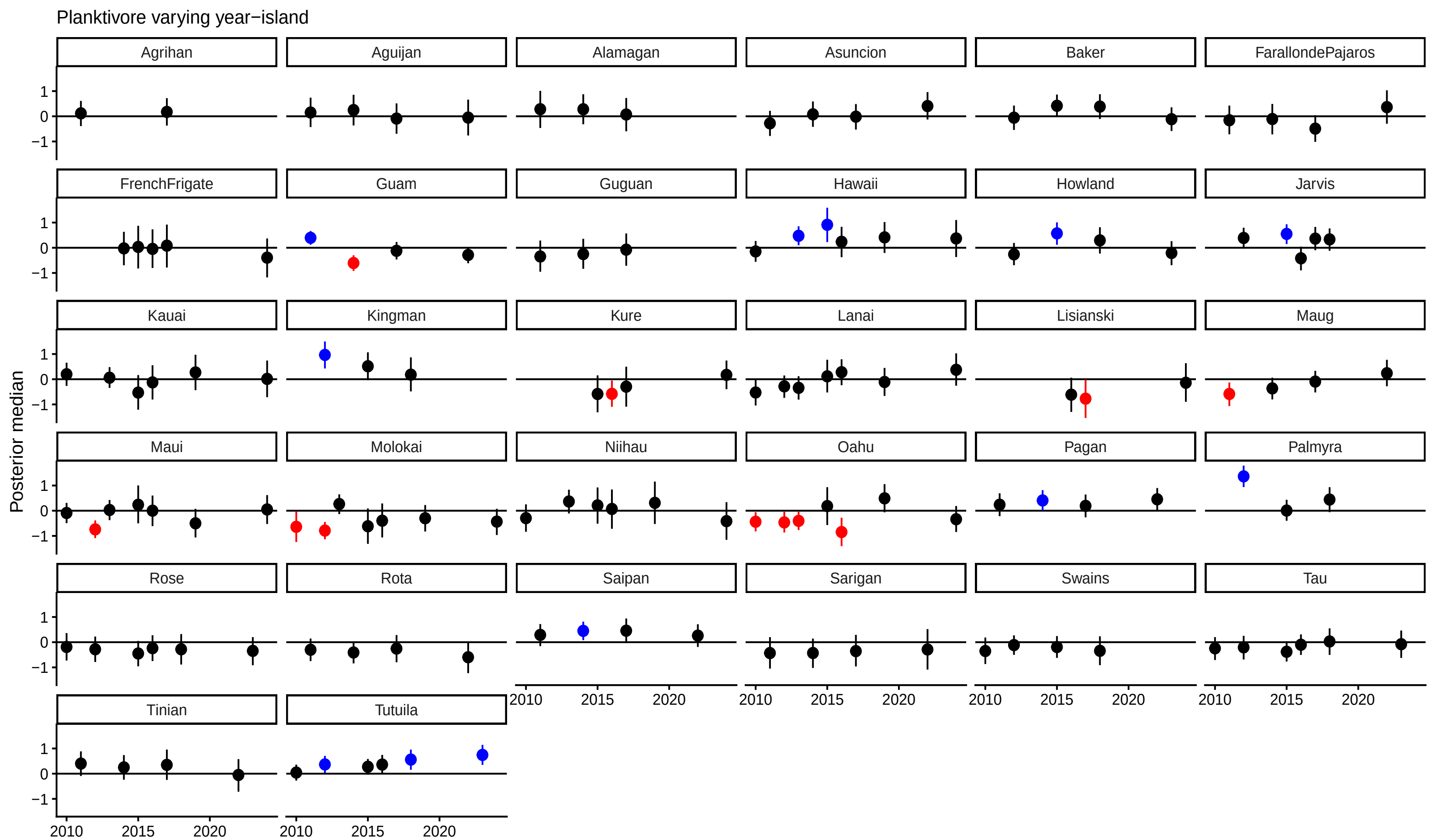

Fig. S7 | Effect of survey year on planktivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).

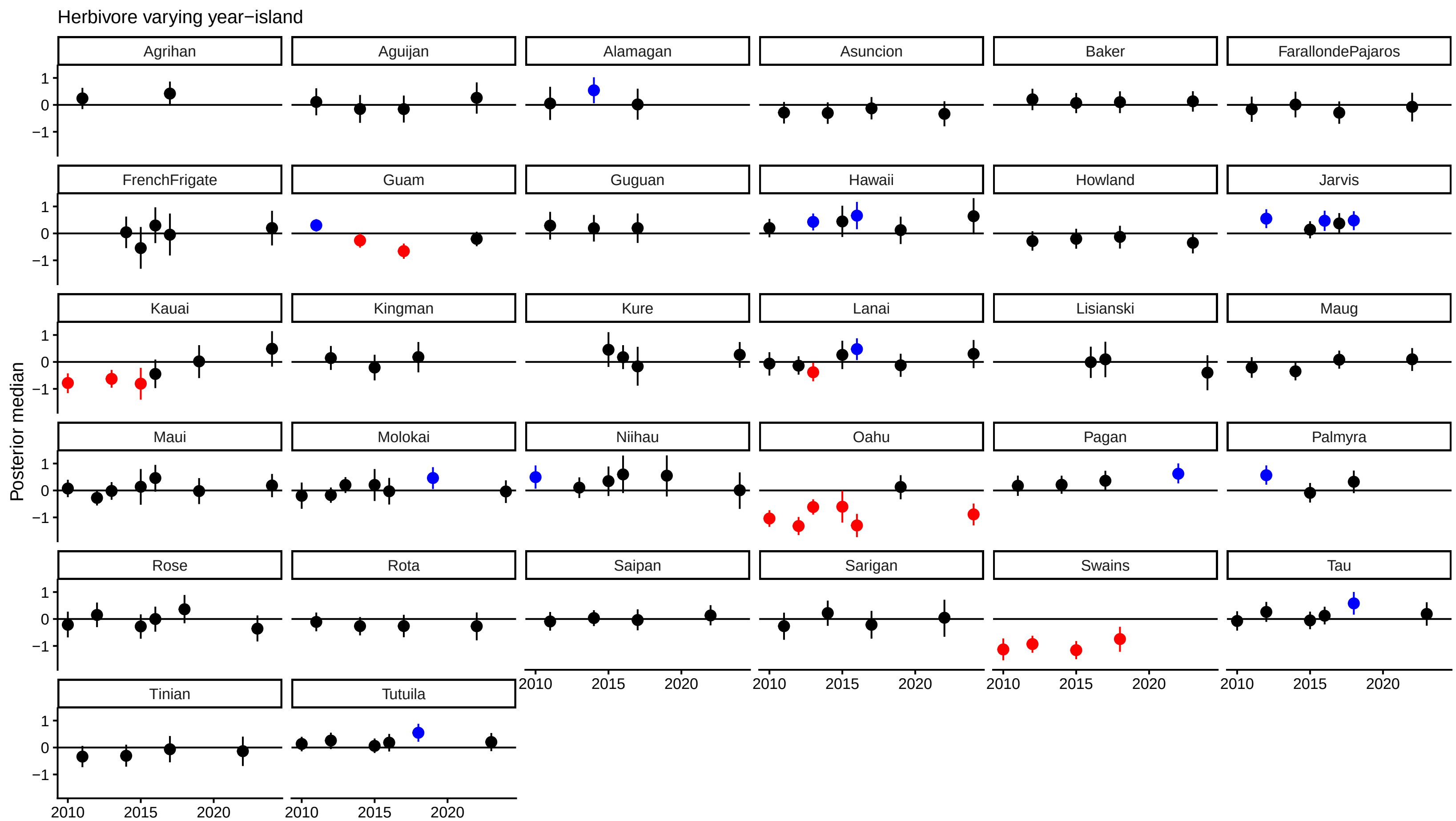

Fig. S8 | Effect of survey year on herbivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).

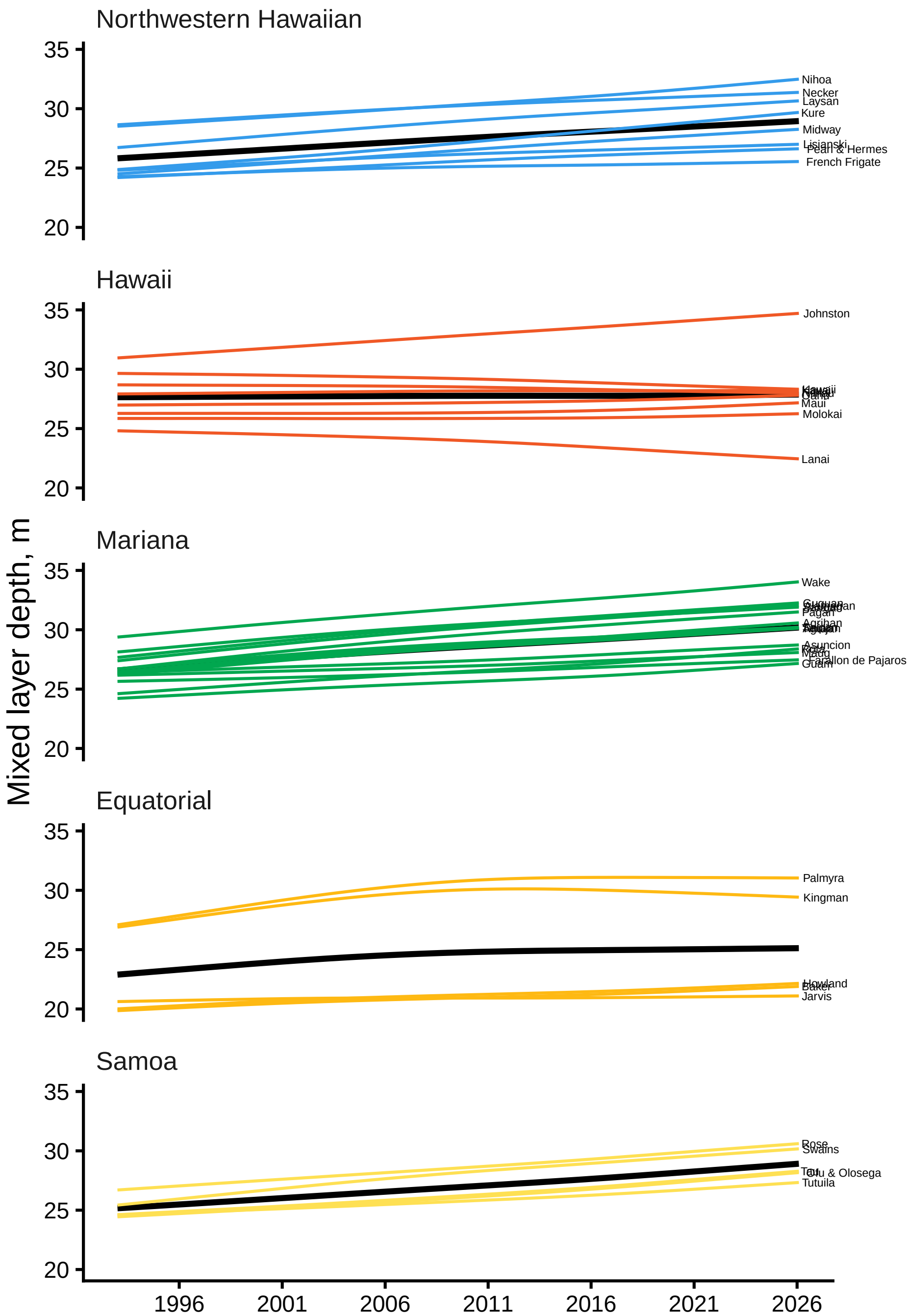

Fig. S9 | Island-level temporal trends in the mixed layer depth layer from 1993-2026. Lines are mean predicted mixed layer, accounting for monthly variability, from a generalised additive model, for each region (black with 95% CI filled by region) and overlaid with mean island predictions (coloured lines).

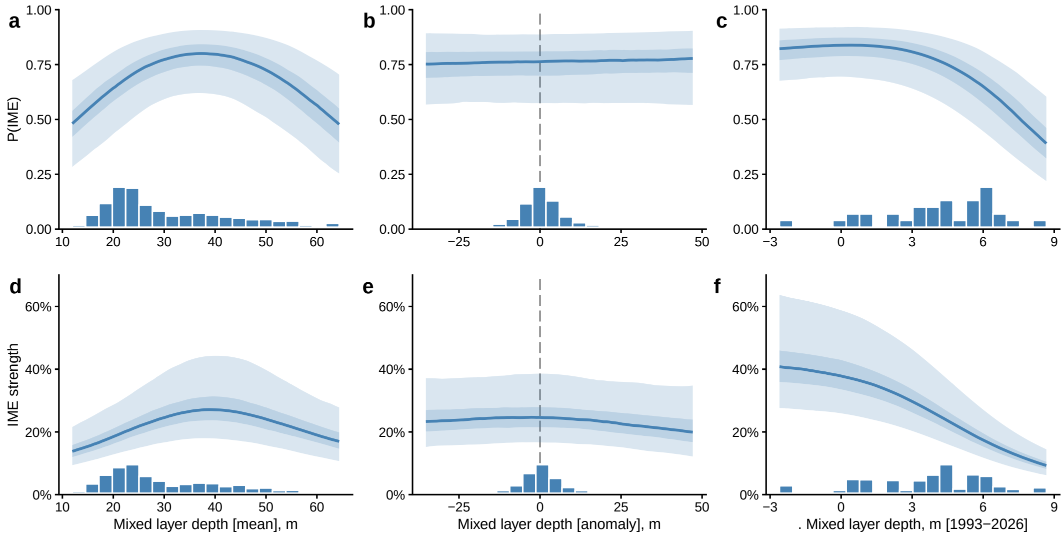

Fig. S10 | Effects of mixed layer depth covariates on IME dynamics from 1997-2025. a) Probability of detecting an IME and b) IME strength (%). Lines are median posterior estimates with 95% and 50% certainty intervals.

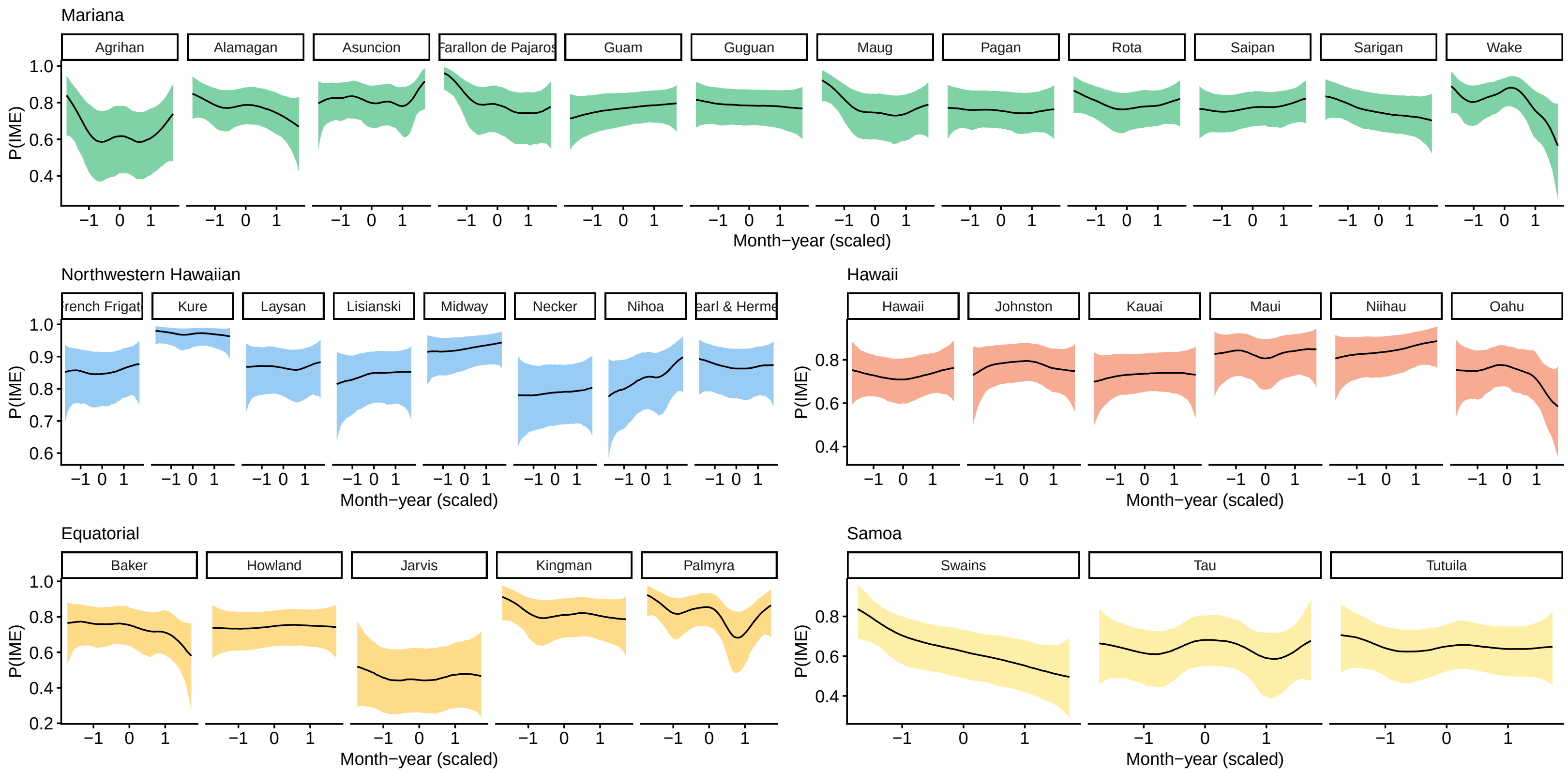

Fig. S11 | Temporal change in the probability of IME occurrence at each island from 1997-2025. Lines are median posterior estimates with 95

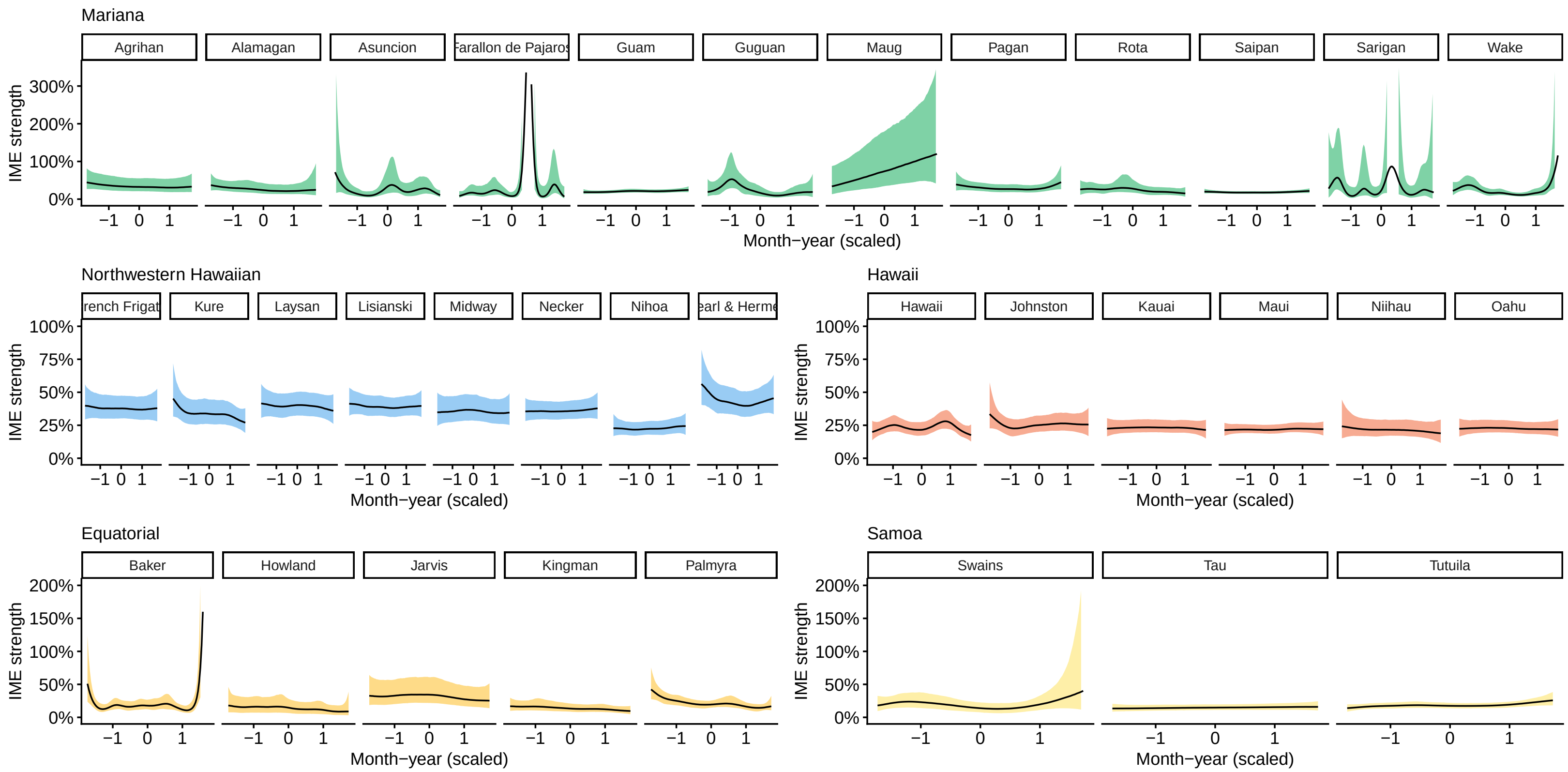

Fig. S12 | Temporal change in the IME strength at each island from 1997-2025. Lines are median posterior estimates with 95

	Covariates	Δelpd	SE(Δelpd)	elpd	SE(elpd)	p_loo	se_p_loo	looic	se_looic
IME detect	**MLD_vars + time + month**	**0.00**	**0.00**	**-7101.91**	**49.58**	**276.47**	**6.64**	**14203.83**	**99.16**
	MLD_vars	-227.59	36.22	-7329.51	33.11	270.96	3.14	14659.01	66.21
	time + month	-361.01	39.64	-7462.92	27.69	206.48	2.04	14925.84	55.38
IME strength	**MLD_vars + time + month**	**0.00**	**0.00**	**5713.96**	**79.43**	**450.04**	**20.96**	**-11427.92**	**158.87**
	MLD_vars	-491.53	41.01	5222.43	74.68	486.53	17.23	-10444.86	149.35
	time + month	-793.65	46.33	4920.30	79.15	460.57	20.12	-9840.61	158.31

Table S1 | Model comparison using leave-out-out cross-validation (LOO). Models fitted to probability of IME detection (IME detect; Bernoulli distributed) and IME strength (chl-a enhancement, %; Gamma-distributed), for nested models with smoothers for the mixed layer climatology (monthly mean), anomaly, and long-term trend of time and month (MLD_vars), and temporal smooths (time + month). Top-ranked models have the lowest Bayesian LOO of expected log pointwise predictive density (elpd). Additional estimates provided for effective number of parameters (p_loo) and LOO information criterion (looic).